WORKED SOLUTION

STEP II 2021, Question 5

Pure • First-Order Differential Equations

STEP II 2021, Question 5

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2021 STEP papers (a zip file):

www.ocr.org.uk/Images/696588-2021-past-papers.zip

The key idea

Part (i) is a first-order equation that is neither separable nor obviously linear in a friendly form, but the suggested substitution turns it into one we can integrate directly. Part (ii) looks geometric, yet once we write down the equation of the tangent and insist that it passes through $(1, t)$, we get exactly the equation of part (i) with $a = 1$. The sketches then come from the derivative, which turns out to be remarkably simple.

Solution

Part (i)

The equation only makes sense for $x \neq a$, so we work on an interval where $x > a$ or one where $x < a$. With $y = (x-a)u$, the product rule gives

$$\frac{\mathrm{d}y}{\mathrm{d}x} = u + (x-a)\frac{\mathrm{d}u}{\mathrm{d}x}.$$

Substituting into the equation,

$$(x-a)u + (x-a)^2\frac{\mathrm{d}u}{\mathrm{d}x} = (x-a)u - x \quad \Longrightarrow \quad \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{x}{(x-a)^2}.$$

The u terms cancel, which is the whole point of the substitution. Writing $x = (x-a) + a$,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x-a} - \frac{a}{(x-a)^2}, \qquad \text{so} \qquad u = -\ln|x-a| + \frac{a}{x-a} + c$$

for a constant c. Multiplying by $x - a$,

$$\boxed{y = a + (x-a)(c - \ln|x-a|)}$$

where the constant c may differ on the two sides of $x = a$.

To check: $y' = c - \ln|x-a| - 1$, so $(x-a)y' = (y-a) - (x-a) = y - x$, as required.

Part (ii)

The tangent at $(t, f(t))$ is $y = f(t) + f'(t)(x - t)$. It passes through $(1, t)$ exactly when

$$t = f(t) + f'(t)(1 - t), \qquad \text{that is,} \qquad (t - 1)\, f'(t) = f(t) - t.$$

Renaming t as x, this is the equation of part (i) with $a = 1$, holding for $x \neq 1$. Hence on each of $x < 1$ and $x > 1$,

$$f(x) = 1 + (x - 1)(c - \ln|x - 1|),$$

with a constant that we fix separately on each side.

? **Ask yourself** Why can't one condition, such as $f(0) = 0$, fix the curve for all x?

The equation tells us nothing at $x = 1$, so the solution on $x < 1$ and the solution on $x > 1$ are independent. That is why the question gives a separate condition for each part.

(a) For $x < 1$, $|x - 1| = 1 - x$. The condition $f(0) = 0$ gives $0 = 1 + (-1)(c - \ln 1) = 1 - c$, so $c = 1$ and

$$\boxed{f(x) = 1 + (x - 1)(1 - \ln(1 - x)) \quad (x < 1).}$$

Differentiating, $f'(x) = 1 - \ln(1 - x) + (x - 1) \cdot \dfrac{1}{1 - x} = -\ln(1 - x)$. So:

- $f'(x) = 0$ only when $1 - x = 1$, giving the single stationary point $(0, 0)$. Since $f''(x) = \dfrac{1}{1 - x} > 0$ for $x < 1$, the curve is convex and $(0, 0)$ is a minimum. So $f(x) \geq 0$, and the curve just touches the x-axis at the origin.
- As $x \to 1^-$, put $z = x - 1 \to 0$: $f(x) = 1 + z - z \ln|z| \to 1$, using $z \ln|z| \to 0$. The gradient $-\ln(1 - x) \to +\infty$, so the curve heads towards $(1, 1)$ and becomes vertical there.
- As $x \to -\infty$, both $x - 1$ and $1 - \ln(1 - x)$ are large and negative, so $f(x) \to +\infty$.

Some helpful points: $f(1 - e) = 1$ and $f(-1) = 2 \ln 2 - 1 \approx 0.39$.

(b) For $x > 1$, $|x - 1| = x - 1$. The condition $f(2) = 2$ gives $2 = 1 + 1 \cdot (c - 0)$, so again $c = 1$ and

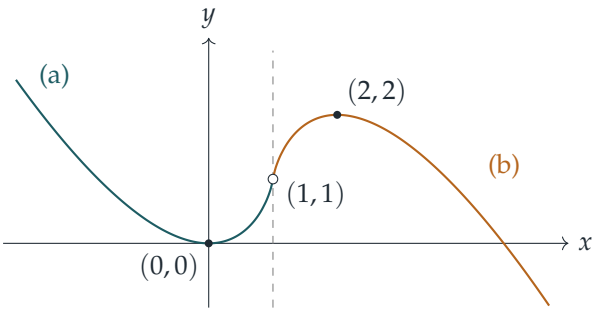
$$f(x) = 1 + (x - 1)\big(1 - \ln(x - 1)\big) \quad (x > 1), \qquad f'(x) = -\ln(x - 1).$$

- The only stationary point is at $x = 2$, namely $(2, 2)$. Here $f''(x) = -\dfrac{1}{x - 1} < 0$, so the curve is concave and $(2, 2)$ is a maximum.
- As $x \to 1^+$, $f(x) \to 1$ (by the same limit) and $f'(x) \to +\infty$: the curve leaves $(1, 1)$ vertically.

- As $x \to \infty$, $1 - \ln(x-1) \to -\infty$, so $\mathrm{f}(x) \to -\infty$. The curve passes through $(1+\mathrm{e}, 1)$ and crosses the x-axis once, near $x \approx 4.59$.

Ask yourself Is there a quicker way to see the shape of (b) once (a) is done?

Yes. Writing $s = |x-1|$, the branch in (a) is $\mathrm{f}(1-s) = 1 - s(1 - \ln s)$, and the branch in (b) is $\mathrm{f}(1+s) = 1 + s(1 - \ln s)$. So $\mathrm{f}(1+s) - 1 = -(\mathrm{f}(1-s) - 1)$: the branch in (b) is the branch in (a) rotated through a half-turn about $(1,1)$. The minimum $(0,0)$ maps to the maximum $(2,2)$.



The curve is not defined at $x = 1$ (the open circle), but both branches approach $(1,1)$ with vertical tangents.

Checks

- Test the tangent property directly. At $t = 0$ the tangent is horizontal, $y = 0$, and it does pass through $(1,0)$. At $t = 2$ the tangent is $y = 2$, which passes through $(1,2)$.
- At $t = -1$: $\mathrm{f}(-1) \approx 0.386$ and $\mathrm{f}'(-1) = -\ln 2 \approx -0.693$. The tangent at $x = 1$ has height $0.386 + 2(-0.693) \approx -1.000$, which equals t.
- The symmetry check: $\mathrm{f}(1+s) + \mathrm{f}(1-s) = 2$ for every $s > 0$, for example $\mathrm{f}(4) + \mathrm{f}(-2) = (4 - 3\ln 3) + (3\ln 3 - 2) = 2$.

Common slips

- Forgetting the modulus in $\ln|x-a|$, and then being unable to handle $x < 1$ in part (a).
- Assuming that the constant found in (a) automatically applies for $x > 1$. Here it happens to come out the same, but that has to be checked using $\mathrm{f}(2) = 2$.
- Concluding that $\mathrm{f}(x) \to 0$ or blows up as $x \to 1$. The term $(x-1)\ln|x-1|$ tends to 0, so $\mathrm{f} \to 1$. It is the gradient that blows up.

Beyond STEP

The equation $(x-a)y' = y - x$ is *homogeneous* once we shift the origin to $x = a$ and $y = a$: with $X = x - a$ and $Y = y - a$ it becomes $XY' = Y - X$. The substitution $Y = Xu$ is then the standard method for homogeneous equations. The point $x = a$, where the coefficient of y' vanishes, is a *singular point*, and the solutions on the two sides of a singular point need not be related at all. Boyce and DiPrima's *Elementary Differential Equations* treats both ideas.

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